MILITARY SPECIFICATION

CLOTH, SATIN, RAYON AND CLOTH, TWILL, RAYON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rayon lining cloths for use in various clothing items.

1.2 Classification. The cloth shall be of the following classes as specified (see 6.1 and 6.2):

Class 1 - Twill, 3.7 ounce
Class 2 - Twill, 4.2 ounce
Class 3 - Satin, 4.5 ounce

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. The dyed and finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material. The yarns shall be multifilament rayon.

3.3 Color. The color shall be as specified (see 6.2) and shall match the standard sample. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed and finished cloth shall show no more labile sulfur than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.4. When dyestuffs are used which require an addition treatment to fix the dyestuffs to the fibers, the supplier shall certify in writing to the contracting officer that in the additive material formaldehyde is not the principle fixing agent and that the additive material used will not create an odor or cause degradation of the cloth during storage.

3.3.1 Matching. The color of the dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness. The dyed cloth shall show fastness to perspiration, wet drycleaning and crocking equal to or better than the standard sample. When no standard sample is established or referenced for colorfastness, the dyed

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cloth shall show not less than "good" fastness to perspiration, wet dry-cleaning, and shall show a Munsell value for crocking no lower than 8.5. When black is specified, the cloth shall show a Munsell value for crocking no lower than 7.5. Testing shall be as specified in 4.4.

3.4 Physical requirements. The physical requirements of the finished cloth shall be as specified in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Class	Weight oz/sq yd (minimum)	Yarns per inch (minimum)		Breaking strength (dry), pounds (minimum)		Breaking strength (wet), pounds (minimum)		Weave
		Warp	Filling	Warp	Filling	Warp	Filling	
1	3.7	121	67	100	50	40	20	2/1 right twill
2	4.2	142	71	115	55	46	22	2/1 right twill
3	4.5	180	67	150	55	60	22	5-harness satin (see fig. 1)

3.4.1 Width. Unless otherwise specified (see 6.2), the width shall be not less than 41 inches inclusive of selvages.

3.5 Finish. The cloth shall be in natural finish. The use of resins, oils, starches or gums in the finishing of the cloth is prohibited.

* 3.6 pH. The pH value of the water extract of the finished cloth shall be no lower than 5.0 nor more than 8.5 when tested as specified in 4.4.

3.7 Shrinkage. The finished cloth shall shrink not more than 6.0 percent in the warp direction or 3.0 percent in the filling when tested as specified in 4.4.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put-up in a roll as specified in PPP-P-1133.

3.9 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

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3.10 Face identification. The face side of the cloth shall be identified by stamping one side with the word "FACE" at each end of the roll.

3.11 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the point level specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the ce-tification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with 4.2.2.1 through 4.2.2.4.

* 4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1. No linear yard (increments of 1 yard on

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the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 38.0 points. If no individual roll exceeds 38.0 points per 100 square yards, the lot shall be acceptable with respect to this characteristic. If one roll exceeds 38.0 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 38.0 points per 100 square yards. For individual roll quality examination, local type defects only shall be scored. Overall type defects shall be scored only in determining compliance with the lot average quality requirements. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Objectionable odor
Width less than specified
Poor dye penetration, mottled, streaky or cloudy
Bias or bowed filling - distortion at any point, 2 or more inches from normal alignment
Turned selvage, which cannot be flattened by manual pressure

4.2.2.2 Examination for length.

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4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Examination for face marking and for compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present:

Face identification missing from either or both ends.

Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3-1/2 continuous yards, full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The sample size (number of sample units) shall be as shown in the table below. The lot shall be unacceptable if one or more sample units fail to meet any requirements specified.

Lot size (yards)	Sample size (units)
800 or less	2
800 up to and including 22,000	3
22,001 and over	5

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TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Rayon identification	3.2	1600 3/
Yarn (multifilament)	3.2	Visual 2/
Labile sulfur	3.3	2020
Fixing agent	3.3	3/
Colorfastness to:		
Perspiration	3.3.2	5680
Wet drycleaning	3.3.2	5622
Crocking	3.3.2	5651
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Dry:		
Warp	3.4	5100
Filling	3.4	5100
Wet:		
Warp	3.4	5100 1/
Filling	3.4	5100 1/
Weave	3.4	Visual 2/
Resins, oils, starches, gums	3.5	3/
pH	3.6	2811
Shrinkage	3.7	4.4.1

1/ Three specimens shall be immersed in tap water at room temperature for 2 hours. The specimens shall then be removed from the water one at a time and tested immediately for tensile strength. The results shall be reported as the average of the three determinations to the nearest 1.0 pound.

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2/ One determination per sample unit, and the result reported as "pass" or "fail".

3/ Unless otherwise specified, a certificate of compliance is required and will be acceptable for the requirements stated.

* 4.4.1 Determination of shrinkage. Shrinkage will be determined in accordance with Method 5558 except that the specimen shall be 22 inches by 22 inches without slits, soaked for 1 hour, and if an extractor is not available the excess water shall be removed by wrapping the specimen in a towel or similar material and squeezing by hand. Wringing by hand or passing the specimen between squeeze rolls is not permitted. The extracted specimen shall be dried by pressing at a temperature of 135° to 149°C.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

* 5.1.1 Level A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirments of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2 and 6.2.1).

* 5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

* 5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES

6.1 Intended use. The specified classes of lining cloth shall be used for the following garments:

Class 1 - General utility use, garrison cap, service cap and hats, male and female and enlisted personnel uniforms.

Class 2 - Enlisted men's coats, male and female officers' and enlisted personnel type uniforms, raincoats with liners.

Class 3 - Officer type overcoats, dress uniforms, (men and women).

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class required (see 1.2).
(c) Color of cloth required (see 3.3).
(d) Width of cloth required, if other than 41 inches (see 3.4.1).
(e) Minimum acceptable length if other than specified (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in PPP-P-1133 may be exercised in the preparation for delivery of this cloth.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 The standard samples have been dyed with the following dyestuff formulations:

Army Green 45

Direct Yellow 105
Direct Yellow 106
Direct Blue 191
Direct Fast Red RL

Gold 68 and Old Gold 175

Direct Blue 191
Direct Yellow 105
Direct Yellow 106

Army Tan 302

Direct Blue 191
Direct Yellow 105
Direct Fast Grey LVL

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 45, 82

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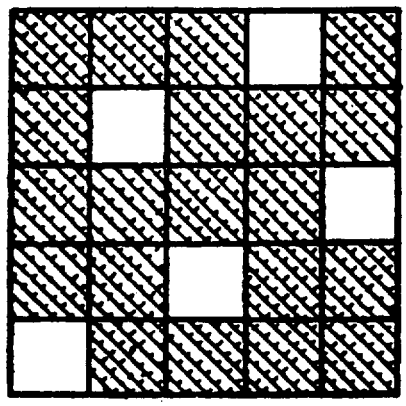
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Class 3

FIGURE 1. *Weave construction.*

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